

Project Overview & Team Responsibilities

1. Project Title: StockWise

Concept: Smart Inventory & Stock Tracking System

Objective: To eliminate the errors and inefficiencies of manual inventory tracking by providing a robust, web-based automated platform for small to medium-sized businesses.

2. Problem Statement

Many businesses struggle with manual stock-taking, leading to data entry errors, overstocking, or unexpected stockouts. This lack of real-time visibility results in financial losses and operational delays.

3. Solution (Project Scope)

StockWise is a web-based management system developed with **Java Spring Boot** and **React**. It offers:

- **Real-time Tracking:** Instant updates on stock levels.
 - **Smart Alerts:** Automated notifications when items reach a critical threshold.
 - **Data Visualization:** Dashboard charts for inventory analytics.
 - **User Management:** Role-based access for Admins and Staff.
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4. Team Members & Roles

Zehra Berre Akyüz B2280.060073 – Project Manager (PM)

- **Primary Responsibility:** Project planning, Jira management, and resource allocation.
- **Key Tasks:** * Defining the project roadmap and 14-week timeline.
 - Monitoring task progress and sprint management on Jira.
 - Handling budget planning and risk management.

Tuba Eraslan B2280.060058 – Backend Lead

- **Primary Responsibility:** Server-side logic, Database management, and API development.
- **Key Tasks:** * Designing the **PostgreSQL** database schema.
 - Developing RESTful APIs using **Java Spring Boot**.
 - Implementing business logic for stock calculations and security.

Ceren Şengül B2380.060018 – Frontend Lead

- **Primary Responsibility:** User Interface (UI) design and Client-side development.
- **Key Tasks:** * Building a responsive web dashboard using **React.js**.
 - Implementing state management and UI components with **Tailwind CSS**.

- Integrating Backend APIs into the frontend using Axios.

Can Tasar B2280.060087 – QA & Documentation Lead

- **Primary Responsibility:** Quality assurance, UML modeling, and technical documentation.
 - **Key Tasks:** * Designing **UML Diagrams** (Use Case & Class Diagrams).
 - Conducting Unit Tests (**JUnit/Mockito**) and Manual Testing.
 - Managing the final project documentation and GitHub repository.
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5. Resource & Budget Summary

- **Human Resources:** 4 Team Members (Allocated 10-12 hours/week).
- **Technological Resources:** * Development: Java, Spring Boot, React, PostgreSQL.
 - Management: Jira, Google Teams.
 - Hosting/Version Control: GitHub.
- **Budget:** 0 TL (Utilizing Open-Source software and academic licenses).